

Demonstrating TapType for mobile ten-finger text entry anywhere

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ABSTRACT

We demonstrate a mobile text entry system that brings full-size ten-finger typing to everyday surfaces, allowing users to type anywhere. Our wearable wristband TapType integrates accelerometers that sense vibrations arising from finger taps against a passive surface, from which our Bayesian neural network estimates a probability distribution over the fingers of the hand. Given a pre-defined key-finger mapping, our text entry decoder fuses these predictions with the character priors of an n-gram language model to decode the input text entered by the user. TapType combines high portability with sustained rapid bimanual input across the full space, which we demonstrate at the example of supplementing text input on mobile touch devices, in eyes-free scenarios using audio feedback, and in a situated Mixed Reality scenario to enable typing outside visual control with passive haptic feedback.

CCS CONCEPTS

• **Hardware** → *Sensor applications and deployments*; • **Human-centered computing** → *Keyboards; Text input; HCI theory, concepts and models*; • **Computing methodologies** → *Bayesian network models*.

KEYWORDS

mobile text entry, invisible interfaces, Bayesian inference, Bayesian neural network, n-gram language model, virtual reality

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1 INTRODUCTION

Physical keyboards are still the most efficient option for entering text, designed to support fast bimanual use through a full-size keyboard layout. In mobile scenarios, however, today's devices do not afford this full-size input, because they compromise on input space and speed for a smaller form factor and thus increased portability. Therefore, in order to detect intended keys soft keyboards on mobile devices have adopted language models which compensate for the input error that comes with smaller layouts [2]. These language models are also useful for text input in Mixed Reality, where the main challenge is not full-size typing, but the lack of typing feedback in mid-air [6]. This can cause fatigue during extended use and prohibits text entry over prolonged periods of time.

In this demonstration, we present a novel text entry system that allows users to opportunistically type on uninstrumented surfaces, using all ten fingers while receiving haptic feedback, which supports the kind of text input they are familiar with from physical keyboards. We detect text input through our prototype TapType [5], which embeds inertial sensors in a wristband to detect touch events on passive surfaces, thereby deriving a probability distribution over the fingers using our Bayesian classifier and fusing them with the character probabilities from an n-gram language model. TapType thus decodes ten-finger text input on everyday passive surfaces in a portable manner, allowing users to type anywhere. TapType supports text input in mobile scenarios that complement touch devices, but is also suitable for eyes-free operation when users receive audio feedback only. The passive surfaces provide support for sustained interaction as well as passive haptic-feedback for rapid input [3].

Furthermore, using TapType, users can transfer their prior experience from ten-finger touch typing on physical keyboards to tap typing on surfaces. This is particularly useful for scenarios in Mixed Reality. Unlike mid-air free-hand text entry with only visual feedback, TapType allows users to type in Mixed Reality as they would on keyboards and laptops. This does not only provide them with passive haptic feedback, which is useful for more engaging experiences in Virtual Reality [1], but also requires no use of cameras or expensive processing for hand tracking. Most importantly, TapType supports text entry outside visual control.

In our full paper [5], we present an end-to-end text entry study of TapType with 10 participants in an online setting. Participants on average typed text at a rate of 19 words per minute and a character error rate (CER) of 0.6%. Expert touch typists consistently achieved an entry rate of more than 25 WPM at a comparably low error rate.

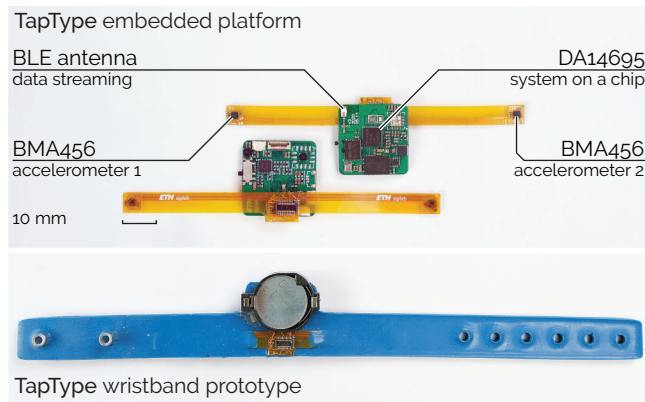


Figure 1: TapType’s wristband integrates two accelerometers and a mainboard in a silicone wrist strap to detect tap events and stream them to a computer.

2 TAPTYPE IMPLEMENTATION [5]

2.1 Hardware

To register tap events, we developed a wireless sensor wristband (Figure 1). Building on our previous prototype TapID [3], we advanced the dual-inertial sensor design with two ultra-low power three-axis accelerometers (BMA456, Bosch Sensortec), which provide higher resolution and higher sampling rates for increased precision. As before, the accelerometers affix to flex PCB boards, which connect to the mainboard. A system on a chip (DA14695, ARM Cortex M33, Dialog Semiconductor) powers our prototypes and streams the data to a PC through Bluetooth Low Energy. A 3 V CR2032 coin cell battery drives TapType for up to 30 hours.

2.2 Input processing

Tap detection: We detect taps by thresholding a running rate-of-change score R_x (adapted from our previous project TapID [3]), accumulating the absolute change in the accelerations’ magnitude over time, while attenuating past accumulations through exponential decay. Contact events cause spikes in R_x above the threshold.

Tap classifier. This key component of our approach is a Bayesian neural network, trained via variational inference to predict the output probability distribution over the fingers. Our use of Bayesian layers allows the classifier to convey better-calibrated *confidence* scores for a given prediction. These probabilities also aid rejecting spurious activations.

Text entry decoder. Using the probability distribution from our Bayesian classifier, we weigh the prior probability from a 12-gram language model for a specific character with the corresponding finger’s probability and find the most likely character sequence through beam search [4]. An additional word-gram model supports our word discovery. We rank all results according to the sum of the log probabilities from the n-gram character language model, the estimated output of the Bayesian neural network as well as the word-gram model, and present them as suggestions to the user.

2.3 Interaction vocabulary

Users type with their fingers as they would on a regular keyboard. For each detected input event TapType plays a click sound and, when using visual feedback, adds an asterisk to the previously entered text. TapType displays the decoded word alongside a list of alternative suggestions from our text entry decoder, replacing the asterisks with the top entry in the suggestion list. The user can confirm a suggestion through entering a space with their right thumb or otherwise step through the list of suggestions by tapping with their left thumb. If the user simply continues typing instead, the list of suggestions disappears and the entered text is replaced by a sequence of asterisks. By tapping with their left palm, the user deletes the current word. The repetition of this gesture deletes previous words subsequently. In case a wrong word was selected, the user can cycle through suggestions for previous words by again tapping with the left thumb. To finish a sentence, the user types space twice, much in the same way as on touch keyboards.

TapType also supports entering arbitrary words by selecting characters one by one, enabling the entry of words that are not part of a dictionary.

3 SCENARIOS POWERED BY TAPTYPE

We present several applications that demonstrate TapType’s suitability to facilitate eyes-free and ad-hoc text entry for different devices and use cases. Our supplementary video shows our system in use during all these scenarios.

Full-size keyboards for smartphones and tablets without occluding screen real-estate. TapType complements text entry on mobile devices in everyday situations, such as mobile phones and tablets. As shown in Figure 2a, the user simply places the phone down and types on the table surface next to it, thereby implicitly increasing the input area for the smartphone’s keyboard. In this way, there is no need to show the keyboard on the smartphone, which frees the entire display for showing content.

Figure 2b shows that the same benefits apply for tablet interaction. Here, users can set up the tablet, focuses their attention on the

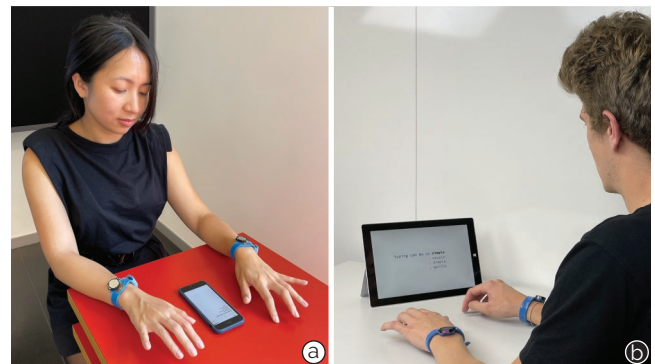


Figure 2: Using the surrounding surface for increased typing convenience, TapType is suitable for text entry (a) on a phone or (b) on a tablet. TapType thus enables setups that are similar to laptop or desktop settings, where users focus their attention on the display while typing in front of it.

screen and type on the table in front of the angled device. Therefore, TapType enables a work setup similar to today's laptops.

Our implementation uses React for a web app that communicates with our text entry backend using WebSockets. The phone app renders the visual feedback that we outlined above, asterisks for words being decoded and suggestions laid out above the text field.

Audio-only feedback for typing on the go. For many existing text entry systems users type under visual control. TapType additionally supports a mode that may be particularly beneficial in mobile and ad-hoc settings and that affords eyes-free text input with only audio feedback (Figure 3). Here, the system acknowledges tap events by playing a click and reads out the most likely suggestion as it appears, as well as the subsequent suggestions as the user steps through them. After selecting the intended word, TapType reads out the entire phrase. Similarly, it plays a tone when the user deletes previously entered words, and recites the remaining text.

We implemented the audio playback to run asynchronously with playback speed set to 1.5× using the *sounddevice* library for Python. A previous playback is interrupted as soon as new text is read out.

Text entry in situated Mixed Reality. We also show the use of TapType for immersive scenarios in Mixed Reality. A particular benefit of our system is in situated Virtual Reality scenarios where efficient text entry is an important area of research (Figure 4). Similar to the other scenarios, TapType continues to take advantage of users' prior experience in touch typing and thus their mental model of text input, which enables a keyboard-free text entry interface in VR. As an input aid, our app can selectively display the corresponding groups of letters above each user's finger.

We implemented the VR scenario using Unity. We render the user's hands according to the tracking information provided by the Oculus Quest 2. Input events and key presses are triggered by



Figure 3: TapType affords eyes-free text entry, which is useful in mobile and ad-hoc situations. Our audio feedback system reads out typed words, suggestions, and sentences.



Figure 4: In situated Mixed Reality, here a replica of an office with virtual monitors, TapType complements typing with passive haptic feedback for productivity scenarios such as writing an e-mail.

TapType's processing pipeline, which handles the event detection, finger classification and text decoding.

As TapType detects touches outside the headset's field of view, it augments text input for today's VR headsets that rely on camera-based input detection. This lifts any constraints on the users' viewing angle, allowing them to focus their attention anywhere in the 3D graphical user interface while entering text.

TapType is not limited to Virtual Reality, but also suitable for Augmented Reality scenarios. TapType could enhance productivity scenarios in future office spaces by enabling opportunistic and ad-hoc text entry on passive surfaces such as tables.

4 CONCLUSION

We presented a demonstration of TapType, a novel method to decode text from typing events on passive surfaces. Our prototype integrates inertial sensors in a flexible wristband, from which we obtain finger tap events through a novel Bayesian neural network. The key enabler of our method is a text entry decoder that takes the probabilistic output of the network as input and fuses it with the likelihood estimate from an n-gram language model. TapType leverages users' ability to transfer their prior experience from ten-finger touch typing on physical keyboards to *tap typing* on surfaces. We demonstrated TapType in scenarios that involve mobile text entry to complement smartphones and tablets as well as for ad-hoc situations using audio feedback only. TapType supports interaction in mixed reality, particularly situated virtual reality, where our approach allows users to enter text *outside visual control*. Taken together, we believe that TapType is a promising enabler for a future generation of wearable and mixed reality systems to type anywhere. For full details of our system and its evaluation, please see the paper [5].

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